

FALL SEMESTER 2026-27
CSE2025 – AWS Solution Architect
Digital Assessment Instruction Manual

Faculty: Dr Renita R

**Design and Deploy a Cloud-Based Application Using Amazon Web Services (AWS) –
(20 Marks)**

Objective

The objective of this assessment is to provide students with practical experience in designing, implementing, and deploying a cloud-based application using Amazon Web Services (AWS). Students are expected to demonstrate their understanding of cloud computing concepts, AWS service selection, cloud architecture design, deployment, security, and best practices. The emphasis of this project is not merely on application development, but on designing an efficient, scalable, and well-architected cloud solution.

Project Description

Each student shall independently design and deploy one cloud-based application based on a real-world use case. Students are free to select any suitable application domain, including but not limited to:

- Student Management System
- Hospital Management System
- Library Management System
- Online Shopping Platform
- Banking Application
- Food Ordering System
- Event Registration Portal
- Portfolio Website
- Blogging Platform
- Chat Application
- Image Storage System
- Inventory Management System
- Online Examination System

- AI-Based Applications
- IoT Monitoring Systems

Students may also propose any other innovative or practical real-world application.

Use Case Selection Policy

To encourage originality and ensure diversity in project work, the following policy shall be followed.

- Every student must work on **one unique use case**.
- **No two students are permitted to work on the same use case.**
- Use cases will be approved on a **first-come, first-approved** basis.
- Once a use case has been approved for a student, it will no longer be available for selection by others.
- Students must obtain approval from the course instructor **before beginning implementation**.
- If a submitted use case has already been approved for another student, the student must select a different use case.

Use Case Approval

Deadline: 04 August 2026

Students must submit the following details for approval:

- Project Title
- Selected Use Case
- Brief Description (100–150 words)
- Proposed AWS Services

Project implementation should commence only after receiving approval from me.

AWS Service Requirements

Mandatory AWS Services

Every project **must** include the following services:

- **AWS IAM** (Identity and Access Management)
- **One Compute Service**
 - Amazon EC2 **or**
 - AWS Lambda **or**
 - AWS Fargate

- **One Storage Service**
 - Amazon S3
- **One Database Service**
 - Amazon RDS **or**
 - Amazon DynamoDB

A minimum of **four AWS services** must be used.

Optional AWS Services

Students may use any additional AWS services that are appropriate for their application.

Examples include:

- Amazon CloudWatch
- Amazon API Gateway
- Amazon SNS
- Amazon SQS
- Amazon CloudFront
- AWS Cognito
- Amazon VPC
- Security Groups
- Route Tables
- Internet Gateway
- Elastic Load Balancing
- Auto Scaling
- Amazon Route 53
- Any other AWS service relevant to the project

Note: The services listed above are optional. Students are **not required** to use all of them. Select only those services that are appropriate for your application's requirements and architecture.

AWS Free Tier Policy

Students are **strongly encouraged** to implement their projects using **AWS Free Tier eligible services**. Some AWS services may incur charges if they are not covered under the AWS Free Tier or if the free usage limits are exceeded. Students who choose to use such services do so entirely at their own discretion. I am **not** be responsible for any AWS charges resulting from:

- Use of paid AWS services
- Exceeding AWS Free Tier limits
- Failure to terminate AWS resources after project completion
- Incorrect AWS configurations
- Resources left running after project completion

Students are advised to:

- Verify AWS pricing before creating resources.
- Monitor their AWS Billing Dashboard regularly.
- Stop or terminate unused AWS resources immediately after completing the project.

Project Deliverables

A. Progress Review Submission

Deadline: 16 September 2026

Students must submit:

- Problem Statement
- Project Objectives
- Proposed AWS Services
- Preliminary AWS Architecture Diagram
- Current Progress Report

This submission will be reviewed to assess project progress and provide feedback before the final evaluation.

B. Final Submission

Deadline: 09 October 2026

Students must submit:

- Final Project Report (PDF)
- AWS Architecture Diagram
- Complete Source Code
- AWS Console Screenshots
- Deployment Evidence

C. Deployment Demonstration (Viva)

Date: 09 October 2026

Students must demonstrate:

- Successful deployment on AWS
- Functional application
- AWS services used
- Complete architecture workflow
- End-to-end execution
- Justification for each AWS service selected

Students should be prepared to answer questions related to their implementation and architectural decisions.

Project Timeline

Activity	Deadline
Use Case Registration & Approval	04 August 2026
Progress Review Submission	16 September 2026
Final Project Report, AWS Architecture Diagram & Source Code Submission	09 October 2026 (A2 Slot) 10 October 2026 (B1, B2 Slot)
Deployment Demonstration (Viva)	09 October 2026 (A2 Slot) 10 October 2026 (B1, B2 Slot)

Evaluation Scheme (20 Marks)

Evaluation Component	Marks
Problem Statement & Objectives	2
AWS Architecture Design	4
AWS Service Selection & Justification	4
Deployment & Demonstration	6
Documentation & Presentation	4
Total	20

Submission Guidelines

Students must ensure that:

- The submitted report accurately reflects the demonstrated project.
- The AWS Architecture Diagram matches the deployed solution.
- The submitted source code is complete and executable.
- Screenshots clearly show the deployed AWS resources.
- The project is fully functional during the demonstration.
- Students are able to explain every AWS service used in their project.

Late Submission Policy

- Students are expected to adhere to all project deadlines.
- Late submissions will attract the following penalties.

Delay	Penalty
Submitted on or before the deadline	No penalty
Up to 3 calendar days after the deadline	20% deduction from the marks awarded for that component
4–7 calendar days after the deadline	50% deduction from the marks awarded for that component
More than 7 calendar days after the deadline	Submission will not be accepted and will be awarded zero (0) marks

- The Deployment Demonstration (Viva) must be completed on the scheduled date.
- Students who fail to attend the demonstration without prior approval from me will receive **zero (0) marks** for the **Deployment & Demonstration** component.
- **Late submission penalties apply independently to each project milestone.**
- Students are advised to submit their work well before the deadline. Delays arising from internet connectivity issues, AWS account problems, Free Tier limitations, billing issues, accidental deletion of resources, forgotten passwords, system failures, or any other technical difficulties **will not** be accepted as valid reasons for extending deadlines or waiving late penalties.
- I have the right to reject any late submission and to make the final decision regarding the evaluation of all project components.

Important Instructions

- **Begin implementation only after receiving approval for your use case.**
- Each student must work on a unique project.
- Use AWS services that best suit your application.
- It is recommended to use AWS Free Tier eligible services wherever possible.
- Maintain proper documentation throughout the project.
- Test your application thoroughly before the final demonstration.
- Keep backup copies of your project files and source code.
- Ensure that AWS resources are terminated after project completion to avoid unnecessary charges.